Solution

SOUND-01

Class 08 - Science

Section A

1.
   (d) greater than

   Explanation: greater than

2.
   (b) Low pitched

   Explanation: A bird makes high pitched sound but a lion makes a low pitched roar. Low pitched sounds are lauder then high pitches sound.

3.
   (c) Number

   Explanation: Shrillness of sound is determined by the number of vibration per second called frequency. Higher the frequency more shrillness is sound.

4.
   (c) Increase

   Explanation: The speed of sound increases with increase in temperature. At higher temperature, the particles of medium have higher energy to vibrate faster.

5.
   (b) Increase

   Explanation: Speed of sound increase with increase in humidity. At higher humidity density of air increases that increase the speed of sound in air.

6. **(a)** 80 dB

   Explanation: The loudness of sound depends upon amplitude of the sound wave. The loudness above 80 dB becomes physically painful noise.

7.
   (c) 20 mm

   Explanation: 20 mm

Section B

8. **(a)** solids, liquids and gases

   Explanation: Sound needs a medium through which it can travel. Sound cannot travel through a vacuum. Sound can travel through solid, liquid, and gases.

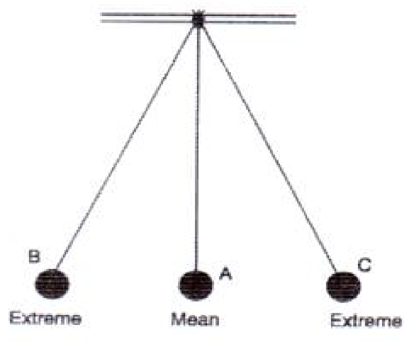
9.
   (c) i, ii & iv only

   Explanation: Because light can travel in vacuum also but it is only sound which requires medium to travel.

Section C

10. No, all bodies cannot produce sound.
11. The loudness of a normal conversation is 60 dB.
12. Flute.
13. Musical sound is one which produces a pleasing sensation.
14. A loud sound is harmful to us. It can damage our eardrum and make us deaf partially or wholly.
15. Yes, while speaking the part which vibrates is called vocal cords. It is below the throat and creates vibrations while speaking.
16. Vibrating part of Flute that produces sound is air column.
17. Dogs can hear frequencies up to 40,000 Hz while bats can produce and hear frequencies up to 70,000 Hz.
18. Television, transistor, radio at high volume, desert coolers and air conditioners are the sources producing noise at home.
19. Sound is the medium by which we communicate with one another.
20. The voice of high frequency is produced.

21. At home, television and transistor at high volumes, some kitchen appliances, desert coolers, air-conditioners all contribute to noise pollution.
22. The most common ill effect of noise pollution is temporary deafness.
23. Presence of excessive or unwanted sounds in environment is called noise pollution.
24. It is characteristic of sound depending upon its frequency. Different frequencies produce different pitch of sound.
25. Jal Tarang works on the principle that the quality of the sound changes with the change in its frequency. Metal tumblers filled with water up to different heights, give vibrations of different frequencies, hence produce different types of sounds.
26. Sound of range 20 hertz to 20,000 hertz are audible range for human ear.
27. These are the two ligaments stretched across the larynx in such a way that it leaves a narrow slit between them for the passage of air.
28. Loudness of sound is expressed in Decibel(db).
29. Sound play an important role in our daily life. Our life depends on sound for each and every action. Without sound we cannot know what others communicate or want to say. Sound enables us to communicate with each other.
30. We feel the vibrations in the bell.
31. Yes, It is the eardrum which vibrates and sends vibrations to the inner ear, when we hear any sound.
32. Sound is produced by vibrating bodies.
33. The rapid linear motion or to and fro motion of a particle or object is called vibration.
34. When amplitude of vibrations is very small, then we cannot see them.
35. The reflected sound if heard distinctly from the original sound is called an echo. It can also be defined as sound heard after reflection from a high wall or obstacle.
36. Examples of Noise: Running of mixer and grinder in the kitchen ,blowing of horns of motor vehicles ,bursting of crackers , barking of dogs , shattering of glass ,landing and flying of aeroplane ,sounds coming from construction site, all students start talking together loudly in a classroom etc.
    Examples of musical sound: All the musical instruments produce musical sounds.The speakers of radio, stereo system and television also produce musical sound.
37. Guitar and sitar are the two musical instruments that produce sound by vibrating strings.
38. The voice of high frequency is produced when the vocal cards are tight and thin.
39. The maximum displacement of any object from its mean position during oscillations is known as amplitude. It is expressed in metres.
40. The vocal cord of a man is about 20 mm long.
41. Those instruments have taut strings, which vibrate when they are plucked, struck, or played with a bow.
42. Presence of unwanted gases and dust particles in air is called air pollution.
43. Loudness of sound is determined by its amplitude.
44. The to and for motion of the object is called vibration. This motion in either side of the object from its mean position is also called the oscillatory motion.



45. Velocity is nothing but Speed with respect to Direction. It is the distance travelled by sound waves in unit time. The velocity of sound at 20°C temperature is considered approximately 340 m/sec.
46. (i) To grow more and more plants along the road sides.
    (ii) We should not play radio ,stereo systems and televisions too loudly.
    (iii) The horns of motor vehicles should not be gone unnecessarily.
    (iv) The bursting of crackers should be avoided.
47. No, We cannot feel vibrations.
48. These are the types of waves which need no medium to travel. Radio waves have the longest wavelengths in the electromagnetic spectrum. These waves can be longer than a football field or as short as a football. Radio waves do more than just bring music to

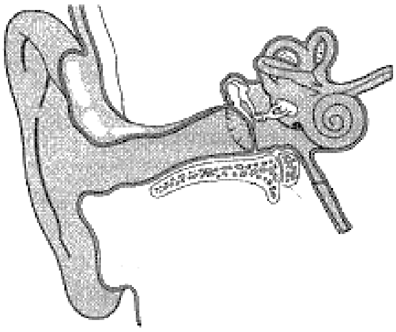
your radio.

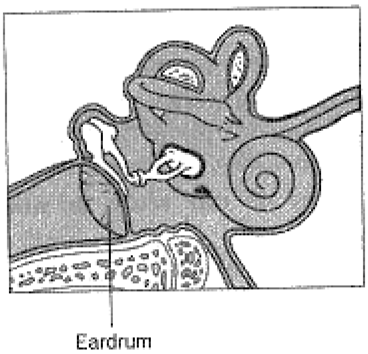
49. Initially, the person is able to hear the sound coming from air and water distinctly. But after some time, when the air is completely removed from the bottle, the sound will pass through the water and not reached to man. So, the man will not hear the sound which was coming through the air initially.
50. Five sources of noise are:
    (i) Noise is produced by traffic.
    (ii) Noise produced by coolers, mixer-grinder etc.
    (iii) Loud sound of TV, radio etc.
    (iv) Loud music in marriages and other religious functions.
    (v) Noise created by construction works in our surroundings.
    (vi) Sound produced by crackers.
    (vii) Use of desert cooler and air conditioners cause noise pollution.
51. Echo is the repetition of sound due to the reflection of original sound by a large and hard obstacle. An echo can be heard only if you are 17 meters away from the surface that reflects sounds. In a small hall distance is less than 17 meter that is why we cannot hear an echo in a small hall.
52. a. i. **Amplitude:** The loudness of a sound depends on its amplitude. If the amplitude of a sound is large, then the sound produced will also be loud.
    ii. **Frequency:** The pitch of a sound depends on its frequency. A sound will be considered a high pitched sound, if its frequency is high.

    b. Sound is a wave motion, produced by a vibrating source. Sound is a form of energy that travels in the form of vibrations through the air or any another medium.
53. a. One vibration in a second is called as Hertz.
    b. To and fro motion of a vibrating object is called oscillation.
    Frequency $= \frac{156}{4} = 39Hz$
54. The number of oscillations made by an object in one second is known as **frequency**. The SI unit of frequency is Hertz.
    The maximum displacement of the object from its mean position is known as **amplitude**. The SI unit of amplitude is meter.
    The time taken to complete one oscillation by an object is called **time period**. The SI unit of time is second.
55. Sounds having frequency more than 20000 Hz are called ultrasounds. Ultrasounds cannot be heard by human beings.
    - Ultrasounds are used in medicines for diagnosing pregnancy, scan internal organs, in the treatment of kidney stones etc.
    - It also detect cracks in metals.
    - It is used to study the growth of foetus inside the mother's womb.
    - It is used in the treatment of muscular pains and a disease called arthritis.
56. The range of human ear is between **20Hz to 20000Hz**.
    The sound frequency between this range is called **audible sound**.
    The sounds having frequency less than 20 Hz is known as **inaudible sounds.**
57. Given that,
    Number of oscillation = 40
    Total time taken = 4 seconds
    Time period = time taken in one oscillation = $\frac{Total\ time}{Total\ number\ of\ oscillation} = \frac{4\ second}{40} = \frac{1}{10}\ second$ = 0.1 second
    Again, frequency = number of oscillations per second = $\frac{Number\ of\ vibration}{Time\ taken} = \frac{40}{4}\ second = 10\ per\ second$ = 10 Hz.
58. We know that speed, pitch, loudness all are initiated with a vibration. During the day, there is a number of vibrations around us. So, the sound coming from the clock gets disturbed and the amplitude of vibrations becomes small. But during the night, there are not such multiple vibrations in the environment. So, the sound is more clear. Further, "the dew factor at night increases the speed of sound as moisture level increases."
59. Bats locate and catch the prey by the mechanism called echolocation. During the hunting time, the bats produce a constant stream of high-pitched sounds. When the sound waves produced by them hits an insect or other animal, the echoes bounce back to the bat that guides them to the prey. The time interval between cry and echo helps to determine the distance of the prey.
60. Excessive or unwanted sounds are called noise. Noise is unpleasant to hear. E.g., Sound produced by a bunch of students speaking together in the classroom.
    The sounds which are pleasing to the ears are called music. It gives a soothing effect rather than creating a chaos in mind. E.g., Sound produced by a harmonium sound.

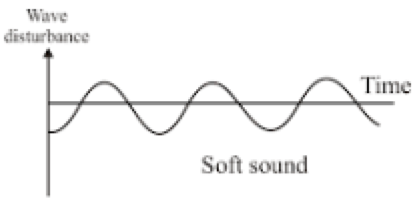
61. One could see lightning during the rainy season before the sound of thunder because speed of light in air is 3×10^8 m/s which is very greater to speed of sound which is around 332 m/s in air. Due to this reason the flash of lightning is seen first and the sound of thunder is heard after a few second.
62. To show that sound travel through liquids, take a bucket full of water and a bell. Now hold the bell gently and dip it into the water bucket, making sure that your hand or bell does not touch the sides of the bucket. Bend your head into the bucket so that your ear touches the surface of water. Gently shake the bell inside the water, you will be able to hear the sound of bell clearly. Thus it is clear that sound waves can travel through liquids.

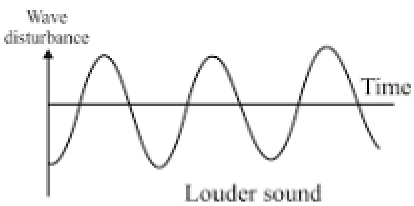
63. Trees have the ability of absorbing sound and not letting it escape and also by reflecting the sound among the trees it lessens the speed of the sound so the impact is generally low. That is why it is advisable to plant trees alongside the roads and around buildings so that it cut down the sound.
64. The distance travelled by a sound wave through a medium is the speed of sound. It can be expressed as-
Speed = Distance/ time
Time of hearing of echo = 4s
Speed of sound = 342 m/s
Distance covered by sound $= 342 \times 4 = 1368m$
Distance of reflecting surface $= \frac{1368}{2} = 684m$
65. Sound is produced due to vibration. The vibration makes the air around vibrate and the air vibrations enter the ear. When an object vibrates, it disturbs the air around it. The molecules of air in contact with vibrating objects also start to vibrate. The vibrating air molecules in turn, make more air, it makes a long chain of sound waves. Sound waves travel in all direction through a material medium like air, water or solids, called propagation of sound. Sound cannot travel in absence of medium because in empty space there is no molecules which help the sound to travel.
66. (i) Ultrasound is used as diagnostic tool in medical science.
(ii) It is used to relieve pains in joints and muscles.
(iii) It is used to detect flaws in metals and structures.
(iv) It is used to test the thickness of various parts.
(v) In the process of electrocardiography, the ultrasonic waves are used to form an image of the heart using reflection and detection of these waves from various parts.
(vi) Medical ultrasound is a diagnostic imaging technique based on ultrasound.
(vii) Ultrasonic waves are used to break stones in the kidney.
67. Humans communicate with each other by speaking. In humans, the sound producing organ is called larynx or voice box which is present at the upper end of the wind pipe in the throat region. Inside the larynx or voice box there are two vocal cords made up of ligaments, which are stretched across it just like a string. When lungs force out air through larynx, the vocal cords vibrate to produce sound. So whenever we speak or sing, our vocal cords vibrate due to the air expelled from the lungs, producing sound.
68. Humans are able to hear with the help of ears. Human ear is sensitive to sound. Human ear consists of three parts i.e. Outer ear, Middle ear and inner ear. The part of ear which we can see is the outer ear which collects sound waves that are conducted through auditory canal. These waves fall on the ear drum and then sent into vibrations. The vibrations of the incoming sound are sent to the next part of the ear. Middle ear enlarges these oscillations. Inner ear converts these pressure waves into electrical signals and send them to auditory nerves. These nerves transfer the signals to the brain and the brain interprets the sound.
69. We hear sound through our ears. The shape of the outer part of ear is like a funnel. When sound enters in it, it travels down a canal at the end of which a thin membrane is stretched tightly. It is called the eardrum. It performs an important function.
The eardrum is like a stretched rubber sheets. Sound vibrations make the eardrum vibrate. The eardrum sends vibrations to the inner ear. From there, the signal goes to the brain. That is how we hear.



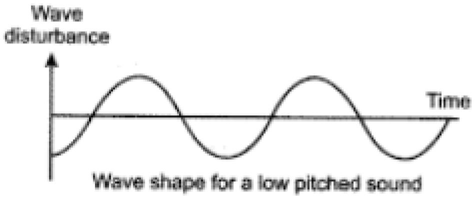


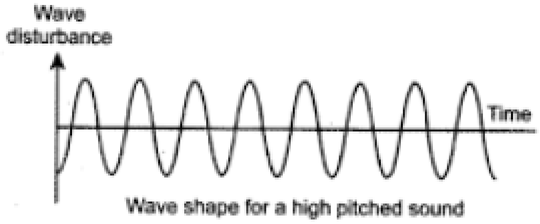
70. Loudness and pitch are both the characteristics of sound. Loudness of sound depends upon its amplitude. A vibration with small amplitude produces a low volume or low loud sound whereas a vibration with large amplitude produces a more loud sound.
Pitch is the shrillness or hoarseness of sound. You often notice that voice of male is heavy and hoarse while the voice of female is sharper. This difference in sound is due to frequency of vibrations. High frequency produces shrill and high pitch sound whereas low frequency produces low pitch sound.
71. Sound is produced due to the to and fro or back and forth motion of an object known as vibration. When a tightly stretched band is plucked, it produces sound. When it stops vibrating, it does not produce sound.
Sound waves enter the ears and travel down a canal at the end of which is a thin, tightly stretched membrane called eardrum. As the sound wave strikes the eardrum, it vibrates and the vibrations reach the inner ear which sends signals to the brain. The brain interprets the signals and we hear the sound.
72. a.





b.





73. Noise pollution is the unwanted and displeasing human created sound that disturbs the environment.

    Effect of noise pollution

    - There are many sources from where noise pollution is created like means of transportation including trains and aircrafts.
    - Setup of industries close to the residential buildings, construction work, voice of loudspeakers etc.
    - The noise pollution affects both health and behavior.
    - It can reduce hearing power of person, become a cause of hypertension, depression and sleep disorders etc.

    Prevention of noise pollution

    - To control noise pollution, the speed of vehicles should be limited, make strict laws to use loudspeakers and any other noise creating tool .
    - Keep the volume of your television under reasonable limits.
    - If you have a pet dog, train it not to bark unnecessarily.
    - If you have a garden area in front or around your house, plant trees and bushes around your house. Not only do they give out fresh air to breathe, they are also known to absorb sound.
74. In **string instruments** like guitar, violin, sitar etc., metal strings of specific metal and thickness are made to vibrate to produce sounds. Sounds of different musical notes can be produced by controlling thickness and tightness of strings in these instrument.

    Percussion instruments are generally used to provide beats to the music e.g. bass drum, dholak, tabla etc. In all these instruments, a specific hollow shape is closed. The vibrated stretched skin produces loud sound by a stretched skin of an animal. The stretched skin is beaten with a stick or by hand to make it vibrate.
75. If there were no voice box in our throat, we would not be able to speak or talk because there is nothing to vibrate through which sound can be produced. In voice box there are two vocal cords. When lungs force out air through voice box, the vocal cords vibrate to produce sound.